DELHI PUBLIC SCHOOL, JANKIPURAM, LUCKNOW

SESSION 2024 -2025; PERIODIC ASSESSMENT-III

MATHEMATICS (041)

SET - II

Max. Marks: 40 Time: 1hr. 30 min.

Date: 16/12/24 Name: Mohd. Mifzaan Class: IX Sec: D

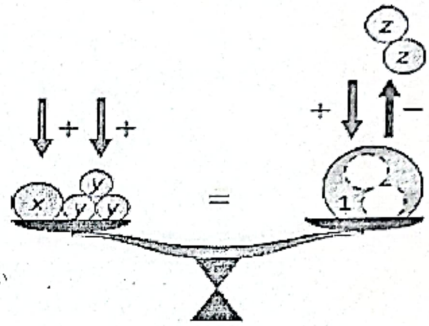
INSTRUCTIONS:

- All sections are compulsory.
- Section A contains 5 MCQs and 2 Assertion Reason based questions carry 1 mark each.
- Section B contains 5 VSA type questions carry 2 mark each.
- Section C contains 3 SA type questions carry 3 mark each.
- Section D contains 2 LA type questions carry 5 mark each.
- Section E contains 1 case study-based question.
- Internal choices are given.

SECTION – A

- **Multiple Choice Questions**

1) A shopkeeper while weighing goods on a physical quantity, gets equivalence of weighs as shown in the figure now obtain the equation from shown figure.



a) $x-3y=1-2z$

b) $x+3y=1-2z$

c) $x+3y=2z$

d) $x-3y=2z$

2) AD is a diameter of a circle and AB is a chord. If AD = 34 cm, AB = 30 cm, the distance of AB from the centre of the circle is:

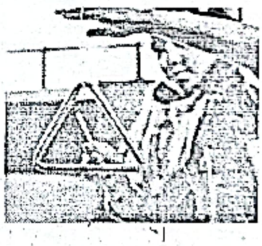
a) 17 cm b) 15 cm c) 4 cm d) 8 cm

3) If E and F are the midpoints of equal sides AB and AC of a triangle ABC. Then:

a) BF = AC b) BF = AF c) CE = AB d) BF = CE

4) Find the area of equilateral triangle of side 6 cm. (by using heron's formula)

a) $9\sqrt{3}cm^2$ b) $2\sqrt{3}cm^2$ c) $4\sqrt{3}cm^2$ d) $6\sqrt{3}cm^2$



5) I) In a ΔABC in which $AB = AC$, the altitude AD bisects BC.

II) If the altitudes AD, BE, and CF of ΔABC are equal, then ΔABC is equilateral.

III) If D is the midpoint of the hypotenuse AC of a right ΔABC then $BD = AC$

Which option is the correct interpretation of above data?

a) I only b) II only c) I and II d) II and III

- **Assertion/Reason**

Directions:

(a) Assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true but reason is not the correct explanation of assertion.

(c) Assertion is true but reason is false.

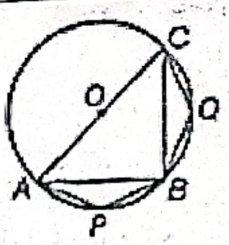
(d) Assertion is false but reason is true.

6) **Assertion(A):** If the angles of a quadrilateral are in the ratio 2: 3: 7: 6, then the measure of angles are 40^0, 60^0, 140^0 120^0, respectively.

Reason(R): The sum of the angles of a quadrilateral is 360^0.

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7) **Assertion:** In the given figure, If O is the center of the circle and AC is the diameter with ∠APB=120°, then ∠BQC=150°.
Reason: In a cyclic quadrilateral, the sum of opposite angles is 180°.



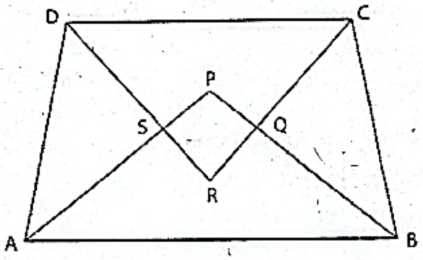
SECTION – B

- **Very Short Answer Type Questions**

8) Find the percentage increase in the area of a triangle if its each side is doubled.

9) The parking charge for vehicles in super Delhi Metro is Rs 20 for first two hrs.' and Rs. 10 for subsequent hr. Assume total parking time to be x hrs. (where x≥ 2) and total parking charge as y. Write the linear equation for above relation. Find the parking charges for 5 hrs. from Graph

10) If bisectors of ∠A and ∠B of a quadrilateral ABCD intersect each other at P, of ∠B and ∠C at Q, of ∠C and ∠D at R and of ∠D and ∠A at S, then prove that opposite angles of PQRS are supplementary.

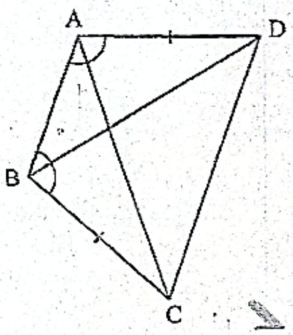


OR

If each pair of opposite sides of a quadrilateral is equal, then it is a parallelogram.

11) D and E are points on side BC of a triangle ABC such that BD=CE and AD=AE. Show that triangle ABD≅ triangle ACE

OR



ABCD is a quadrilateral in which AD = BC and ∠ DAB = ∠ CBA (see in Fig.) Prove that
i) Δ ABD ≅ Δ BAC
ii) BD = AC

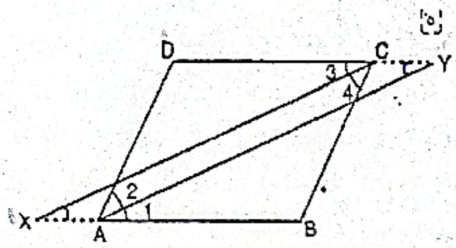
12) **Theorem:** "Measure of angles subtended to any point on the circumference of the circle from the same arc is equal to half of the angle subtended at the center by the same arc."

SECTION – C

- **Short Answer Type Questions**

13) ABCD is a quadrilateral in which the bisectors of ∠A and ∠C meet DC produced at Y and BA produced at X respectively.

Prove that $\angle X + \angle Y = \frac{1}{2}(\angle A + \angle C)$.

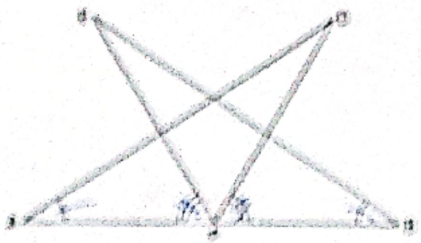


OR

Show that if the diagonals of a quadrilateral bisect each other at right angles, then it is a rhombus.

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14) AB is a line segment and P is its mid-point. D and E are points on the same side of AB such that $\angle BAD = \angle ABE$ and $\angle EPA = \angle DPB$. Show that



i) $\triangle DAP \cong \triangle EBP$

ii) AD = BE

15) If the work done by a body on application of a constant force is directly proportional to the distance travelled by the body, express this in the form of an equation in two variables and also calculate the work done when the force applied is 5 N and distance travelled by the body is:

i) 2 units

ii) 0 unit

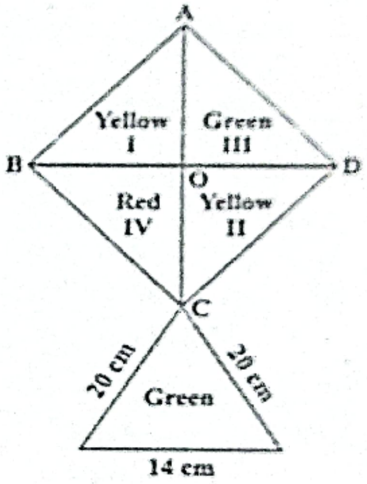
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A fraction becomes 1/4 when 2 is subtracted from the numerator and 3 is added to the denominator. Represent this situation as a linear equation in two variables. Also, find two solutions for this

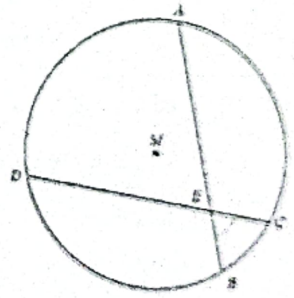
SECTION – D

- **Long Answer Type Questions**

16) How much paper of each shade is needed to make a kite given in Fig. in which ABCD is a square with diagonal 44 cm

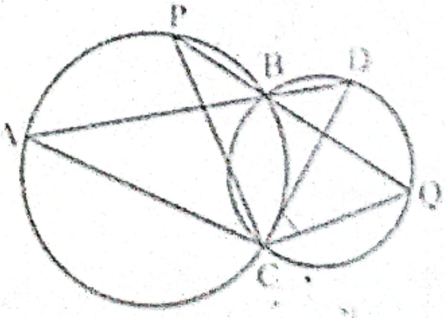


17) If two equal chords of a circle intersect within the circle, prove that the line joining the point of intersection to the center makes equal angles with the chords



OR

Two circles intersect at two points, B and C. Through B, two line segments ABD and PBQ are drawn to intersect the circles at A, D and P, Q, respectively (see in Fig). Prove that $\angle ACP = \angle QCD$.



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SECTION - E

Case Study Based Question

A group of students is studying the properties of triangles. They came across the following scenario: Three friends, Ankit, Bhavna, and Chetan, were discussing their recent hiking trip. During their hike, they noticed a triangular-shaped lake. Their Discussion:

Life Below Water (SDG 14): They observed diverse aquatic life in the lake, highlighting the need for preserving biodiversity and the health of underwater ecosystems.

Life on Land (SDG 15): The trio admired the lush vegetation surrounding the lake, emphasizing the importance of preserving terrestrial ecosystems and preventing deforestation.

Climate Action (SDG 13): The friends discussed the impact of climate change on the lake's water levels and pledged to raise awareness about climate action to protect vulnerable ecosystems like the one they encountered.

Their Observation:

1. The lake has three sides of different lengths.
2. The sum of the lengths of any two sides of the lake is greater than the length of the third side.
3. The largest angle of the lake is less than 90 degrees.
4. The smallest angle of the lake is greater than 30 degrees.

i. The type of triangle formed by the lake is:

(a) Equilateral (b) Isosceles

(c) Scalene (d) Right-angled

ii. The sum of the measures of the three exterior angles of the lake's triangle is:

(a) 180 degrees (b) 90 degrees

(c) 360 degrees (d) It cannot be determined

iii. The lake's triangle is an example of a triangle that satisfies the:

(a) Angle-side-angle (ASA) condition (b) Side-angle-side (SAS) condition

(c) Side-side-side (SSS) condition (d) None of the above

iv. The measure of the largest angle of the lake's triangle is:

(a) 30 degrees (b) 60 degrees

(c) 90 degrees (d) More than 90 degrees

OR

The measure of the smallest angle of the lake's triangle is:

(a) 30 degrees (b) 60 degrees

(c) 90 degrees (d) Less than 30 degree

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